MA1512 – Differential Equations for Engineering

A Book of Problems

Fall 2026

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How to Use This Book

This book is a collection of problems to help you develop the computational techniques of differential equations, as well as deepen your understanding of the ideas behind them.

Before you embark on the course, take a look through the first section, **Before We Begin**: this section presents problems in calculus, the hyperbolic functions, and the algebra of complex numbers that involve skills you will rely on for the next few weeks (and beyond). Work through the problems here first, and feel free to come back to them should a later problem trip you up on a prerequisite.

Each of the remaining sections corresponds to a lecture and contains three courses.

Appetizers. These are short drills that will train you on basic techniques. You should be able to work through these problems immediately after attending each lecture. If you encounter any trouble, revisit the corresponding section in *A Book of Examples*.

Main Course. These problems will be somewhat longer, often combining various techniques, involving more challenging computations, or requiring some form of mathematical modeling. Expect to spend more time here. Your tutor will walk you through these problems in class, so do attempt them beforehand.

Room for Dessert. These are *optional* problems for those who want a challenge. The problems will *not* be tested on the exam, but they're where some of the more interesting ideas in the course appear.

Hints and answers to selected problems are presented at the end of the book.

Before We Begin

Calculus of a Single Variable

1. Use implicit differentiation to find the slope of the tangent line to each of the following curves at the given point.
   (a) astroid: $x^{2/3} + y^{2/3} = 4$ at $\left(-3\sqrt{3}, 1\right)$
   (b) cardioid: $x^2 + y^2 = \left(2x^2 + 2y^2 - x\right)^2$ at $(0, 1/2)$
   (c) conchoid: $x^2y^2 = (y+1)^2\left(4 - y^2\right)$ at $\left(2\sqrt{3}, 1\right)$
2. Evaluate the following integrals.

   (a) $\displaystyle\int e^{\sqrt{x}}\,dx$

   (b) $\displaystyle\int \frac{1}{x\sqrt{\ln x}}\,dx$

   (c) $\displaystyle\int \sqrt{\frac{1-x}{1+x}}\,dx$

   (d) $\displaystyle\int t\sin t\cos t\,dt$

   (e) $\displaystyle\int_1^3 r^4 \ln r\,dr$

   (f) $\displaystyle\int_0^4 \frac{x-1}{x^2-4x-5}\,dx$

   (g) $\displaystyle\int \frac{x-1}{x^2-4x+5}\,dx$

   (h) $\displaystyle\int \sin^5 t\cos^4 t\,dt$

   (i) $\displaystyle\int_1^4 \frac{e^{\sqrt{t}}}{\sqrt{t}}\,dt$

   (j) $\displaystyle\int_0^4 \frac{6z+5}{2z+1}\,dz$

The Hyperbolic Functions

1. (a) Show that $\cosh^2 x - \sinh^2 x = 1$.*
   (b) From the previous result, show that $\operatorname{sech}^2 x + \tanh^2 x = 1$.
2. (a) Show that $\dfrac{d}{dx}\tanh x = \operatorname{sech}^2 x$.
   (b) Show that $\displaystyle\int \tanh x\,dx = \ln(\cosh x) + c$, where $c \in \mathbb{R}$.
3. The inverse hyperbolic cosine function is given by

$$y = \cosh^{-1} x \quad \Leftrightarrow \quad \cosh y = x, \quad y \geq 0,\ x \geq 1.$$

   Use implicit differentiation to show that, for $x > 1$,

$$\frac{d}{dx}\cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}.$$

* This is in contrast to $\cos^2 x + \sin^2 x = 1$.

The Complex Numbers

1. Let $z = 2 - 3i$ and $w = 1 + 2i$. Express each of the following complex numbers in the form $a + ib$, where a and b are real numbers.

   (a) zw (c) z/w

   (b) w/z (d) $z^3 - (z^*)^3$

2. (a) Express $z = 1 + i$ in exponential form.

   (b) Express $w = e^{-i\pi/6}$ in Cartesian form.

3. (a) Use Euler's formula to show that, for all $\theta \in \mathbb{R}$,

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

   (b) We may take the formulae from the previous part as the definitions of $\cos z$ and $\sin z$ for any $z \in \mathbb{C}$. Show that, for $x \in \mathbb{R}$,

$$\cos(ix) = \cosh x, \quad \sin(ix) = i\sinh x.$$

   Similarly, show that

$$\cosh(ix) = \cos x, \quad \sinh(ix) = i\sin x.$$

   (c) Hence, or otherwise, show that

$$\tan(ix) = i\tanh x.$$

4. Evaluate $z = i^{-i}$.

   Hint: Use Euler's formula!

5. The Maclaurin series expansion of e^x is given by

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots.$$

   (a) Write out the first few terms of the Maclaurin series expansion of $e^{i\theta}$. (You may do this by replacing x with $i\theta$ in the expansion of e^x.)

   (b) Hence, or otherwise, determine $\mathfrak{Re}\left(e^{i\theta}\right)$ and $\mathfrak{Im}\left(e^{i\theta}\right)$.

   (c) Write down the Maclaurin series expansions of $\cos x$ and $\sin x$ about $x = 0$. How does this compare to your results from the previous part?

An Introduction to Ordinary Differential Equations

Appetizers

1. For each of the following differential equations: (i) determine any equilibrium solutions and classify their stability, (ii) sketch a direction field, and (iii) determine the behavior of the function $y(x)$ as $x \to \infty$: does this behavior vary depending on the value of $y(0)$?

   (a) $y' = y + 1$

   (b) $y' = 2y - 2$

   (c) $y' = y^2 - 4$

2. Solve the following first-order differential equations.

   (a) $x(x+1)\,y' = 1$

   (b) $\sec(x)\,y' = \cos 5x$

   (c) $(1+y)\,y' + (1-2x)\,y^2 = 0$

3. Consider the first-order differential equation

$$\frac{dy}{dx} = (x - y + 1)^2.$$

   (a) Let $u = x - y + 1$. Express dy/dx in terms of x, u, and du/dx, and rewrite the above equation in terms of the variables x and u.

   (b) Solve the differential equation in x and u by performing a separation of variables. Hence, determine a general solution of the original differential equation by substituting back $u = x - y + 1$.

   (c) Does your solution work for all values of x and y? Does the differential equation have any other solutions in addition to the general solution you obtained?

4. Consider the first-order differential equation

$$\frac{dy}{dx} = \frac{y - 4x}{x - y}.$$

   (a) Let $v = y/x$. Express dy/dx in terms of x, v, and dv/dx, and rewrite the above equation in terms of the variables x and v.

   (b) Solve the differential equation in x and v by performing a separation of variables. Hence, determine a general solution of the original differential equation by substituting back $v = y/x$.

   (c) Does the differential equation have any other solutions in addition to the general solution you obtained?

5. If we start with an initial concentration C_i of radioactive radon-222, what fraction of the original concentration remains after 5 days? The half-life for radon is 3.8 days.

6. The bacteria in a certain culture number 10000 initially. Two and a half hours later, there are 11000 of them. Assuming a Malthusian (exponential) model for the growth of bacteria, determine the number of bacteria 10 hours after the start of the experiment. How long will it take for the number to reach 20000?

Main Course

1. Employ an appropriate substitution to transform the following equations into separable first-order differential equations, and hence solve the initial value problem.

   (a) $y' = \left(\dfrac{x+y+1}{x+y+3}\right)^2$, with $y(0) = 0$.

   (b) $xe^{y/x} + y = xy'$, with $y(1) = 0$.

   (c) $y'y'' = 2$, with $y(0) = 1$ and $y'(0) = 2$.

2. (a) The inverse hyperbolic sine function is given by

$$y = \sinh^{-1} x \quad \Leftrightarrow \quad \sinh y = x \text{ for } x \in \mathbb{R}.$$

   Use implicit differentiation to show that

$$\frac{d}{dx} \sinh^{-1} x = \frac{1}{\sqrt{1+x^2}}.$$

   (b) If a cable is held up at two ends at the same height, it will sag in the middle, making a U-shaped curve called a *catenary*. It can be shown using principles of physics that the shape is given by a function $y(x)$ that satisfies the *second-order* differential equation

$$\frac{d^2y}{dx^2} = \frac{\mu}{T}\sqrt{1+\left(\frac{dy}{dx}\right)^2},$$

   where μ is the weight per unit length of the cable, and T is the horizontal component of its tension, both of which are constant. The coordinate system is chosen such that $x = 0$ at the lowest point of the catenary, and $y(0) = 0$.

   Employ an appropriate substitution to transform the equation into a separable first-order differential equation. Hence, or otherwise, determine the function $y(x)$ that describes the shape of the cable.

3. A tank contains 100 liters of water and an unknown amount of salt. Water containing 0.25 kilograms of salt per liter flows into the tank at a rate of 3 L sec^{-1}, and the well-mixed salt solution flows out of the tank at the same rate so the amount of liquid in the tank remains constant.

   (a) Write down a differential equation for the amount of salt in the tank at any time. You may choose your own symbols and units. *Do not solve this differential equation.*

(b) Determine the equilibrium solution(s) of your differential equation. How much salt will the tank's mixture contain after a very long time? Does this amount depend on the amount of salt initially present?

4. The population of a certain species of bugs behaves according to the Verhulst (logistic) model:

$$\frac{dy}{dt} = k\left(1 - \frac{y}{y_\infty}\right)y,$$

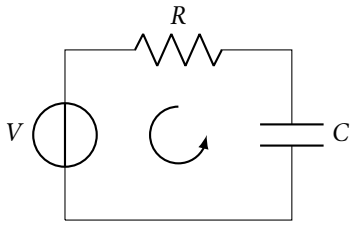
with growth constant $k = 1.5$ per day when the population is small. After some time, the bug population stabilizes to a carrying capacity of $y_\infty = 375$ bugs. You wish to start testing out a new insecticide on the bugs you have collected.

(a) What is the maximum number of bugs you can put to death per day without causing the population to die out?

(b) Suppose you remove 80 bugs per day from the bottle, and that all of these bugs die as a result of being sprayed with this insecticide. How many bugs will you have in the long run?

Room for Dessert

1. Consider a simple circuit containing a capacitor, resistor, and battery, as shown in the figure below.



From Kirchhoff's Laws, it can be shown that the charge $Q(t)$ on the capacitor satisfies the first-order differential equation

$$R\frac{dQ}{dt} + \frac{Q}{C} = V.$$

This equation contains the constants

- R, resistance measured in ohms (Ω),
- C, capacitance measured in farads (F), and
- V, voltage supplied by the battery measured in volts (V).

(a) Determine the equilibrium solution(s) of the differential equation.

(b) Suppose that $Q(0) = 0$. Find the particular solution $Q(t)$, and sketch the graph of Q against t. Does this agree with the equilibrium solution you obtained from the previous part?

(c) Suppose that, at time $t = t^*$, the battery is removed and the circuit closed once again. Find $Q(t)$ for $t > t^*$.

2. Consider the following first-order *linear* differential equation:

$$y' + p(x) \cdot y = q(x).$$

Here, $p(x)$ and $q(x)$ are some functions in the independent variable x.

(a) Let $q(x) = 0$ be the constant zero function, so that we have the differential equation

$$y'(x) + p(x) \cdot y(x) = 0.$$

Show that a solution is given by

$$y = Ae^{-\int p(x)\,dx}, \text{ for some } A \in \mathbb{R}.$$

(b) Suppose, now, that $q(x)$ is not the constant zero function:

$$y' + p(x) \cdot y = q(x).$$

Suppose it is known that a solution is given by

$$y = A(x)\, e^{-\int p(x)\,dx},$$

where $A(x)$ is some function of x. By substituting the suggested solution into the differential equation, show that

$$A(x) = \int q(x)\, u(x)\; dx, \quad \text{where} \quad u(x) = e^{\int p(x)\,dx}.$$

Linear Differential Equations

Appetizers

1. Solve the following first-order linear differential equations.

   (a) $x y' + (1 + x)\, y = e^{-x}$

   (b) $y' + y \cos x = e^{-\sin x}$

   (c) $y' - \left(1 + 3x^{-1}\right) y = x + 2$

2. Find a general solution for each of the following homogeneous, second-order linear differential equations, and hence determine the particular solution that satisfies the given initial conditions.

   (a) $y'' = 0, \quad y(1) = 2, \quad y'(1) = -1$

   (b) $y'' + 6y' + 9y = 0, \quad y(0) = 1, \quad y'(0) = -1$

   (c) $y'' - 2y' + (1 + 4\pi^2)y = 0, \quad y(0) = -2, \quad y'(0) = 6\pi - 2$

3. The equation of motion of a simple pendulum tied to a string of length ℓ is given by

   $$\ddot{\theta} = -\omega^2 \sin\theta, \text{ where } \omega = \sqrt{g/\ell}.$$

   Here, $\theta(t) \in (-\pi, \pi]$ denotes the angular displacement of the pendulum at time t, and $g = 9.80665 \text{ m s}^{-2}$ is the acceleration due to gravity. When the pendulum is made to oscillate at small angles (i.e., $\theta \approx 0$), we may approximate the sine function as $\sin\theta \approx \theta$.[†] In this case, the equation of motion becomes

   $$\ddot{\theta} = -\omega^2\theta, \text{ where } \omega = \sqrt{g/\ell}.$$

   A simple pendulum of length 1 meter is at rest at its equilibrium position. At time $t = 0$, it is given an initial angular velocity of $\dot{\theta}(0) = 1$ radian per second. Assuming that no other forces act on the pendulum, determine its angular displacement at $t = 0.8$ seconds.

[†] We use the Maclaurin series expansion

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots.$$

Main Course

1. Employ an appropriate substitution to transform the following equations into first-order linear differential equations, and hence find the general solution to the differential equation.

   (a) $2xyy' + (x - 1)\, y^2 = x^2 e^x$

   (b) $y' + y + xy^{-1} = 0$

   (c) $x^2 + \sin y = y' \cos y$

2. Experiments show that the rate of change of the temperature of a small iron ball is proportional to the difference between its temperature, $T(t)$, and that of its environment, T_{env}, which is constant.

   (a) Write down a differential equation describing this situation.

   (b) Solve your differential equation by using an integrating factor.

   (c) The ball is heated to 250°C and then left to cool in a room at 25°C. Its temperature falls to 150°C in half an hour.

      i. Determine the ball's temperature an hour after it begins cooling.

      ii. How long will it take for the ball's temperature to reach 31.6°C?

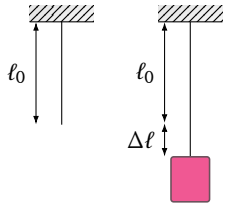
3. For each of the following homogeneous, higher-order linear differential equations: (i) write down its characteristic equation, (ii) find the roots of the characteristic equation, and (iii) write down a general solution for the differential equation.

   (a) $y''' - y'' - y' + y = 0$

   (b) $y^{(4)} - 13y'' + 36y = 0$

   (c) $y^{(8)} - 8y^{(4)} + 16y = 0$

4. A spring of length ℓ_0 meters is attached to the ceiling. A mass of m kilograms is attached to the spring, and then brought to rest, extending the spring by an additional $\Delta\ell$ meters.



   The weight creates tension in the spring, which tries to return to its natural length. Thus, the weight experiences an upward restoring force from the spring. By Hooke's Law,

   > *the restoring force (that returns an object to equilibrium) is directly proportional to the distance of the object from equilibrium.*

   Let $y = 0$ be the position of the mass at equilibrium, and take downward as the positive direction.

   (a) By resolving the forces on the mass at rest, show that

   $$k\Delta\ell = mg,$$

   where k is the spring constant (i.e., the proportionality constant in the restoring force), and g denotes the acceleration due to gravity.

   (b) The mass is pulled downwards by an additional y meters.

      i. Use Newton's Second Law to set up a differential equation describing the vertical motion $y(t)$ of the mass on the spring.

      ii. Show that the mass will exhibit simple harmonic motion about the equilibrium $y = 0$. What is the angular frequency of the oscillation?

iii. If the mass is immersed in a viscous fluid, it experiences an additional drag force with a magnitude proportional to its velocity. Determine the range of values of the damping constant c (i.e., the proportionality constant in the drag force) for which the object will still oscillate about the equilibrium.

Room for Dessert

1. Solve the following initial value problem:

$$3y''' + 5y'' + y' - y = 0, \quad y(0) = 0, \quad y'(0) = 1, \quad y''(0) = -1.$$

2. By Newton's Second Law, the equation of motion of a damped mass-spring system is given by

$$m\ddot{x} + c\dot{x} + kx = 0,$$

where m is the mass of the object, c is the damping constant, and k is the spring constant.

(a) Determine the conditions on the constants c, m, and k such that the system is *underdamped*.

(b) Show that the displacement of the mass is given by a function of the form

$$x(t) = Re^{-\alpha t}\cos(\mu t - \varphi),$$

for some $\alpha, \mu, \varphi \in \mathbb{R}$.

(c) Show that the time between successive peaks of the oscillation is given by

$$T = \frac{2\pi}{\mu}.$$

Hence, show that the ratio of the displacements at two successive maxima x_n^* and x_{n+1}^* is given by

$$\frac{x_n^*}{x_{n+1}^*} = e^{\alpha T}.$$

Note that this ratio does not depend on the pair of successive maxima selected.

(d) The **logarithmic decrement** of the oscillation is defined as

$$\Delta = \ln\left(\frac{x_n^*}{x_{n+1}^*}\right).$$

From the previous part, we can thus conclude that

$$\Delta = \alpha T.$$

Given a physical system, the mass m, damped period T, and logarithmic decrement Δ are quantities that can easily be measured. The logarithmic decrement formula thus gives us a practical method for computing the damping constant of a system, which is often more difficult to measure directly.

In a certain experiment, the displacement of a 1.5 kg mass on a spring is measured. It is observed that the mass moves from a peak at $x_1^* = 4$ cm to the next peak at $x_2^* = 3.2$ cm in 1.5 seconds. Use the logarithmic decrement formula to determine the damping constant c.

The Harmonic Oscillator

Appetizers

1. Find the general solutions to the following non-homogeneous linear differential equations by employing the method of undetermined coefficients.

   (a) $y'' + 2y' + 10y = 25x^2 + 3$

   (b) $y'' - 6y' + 8y = x^2 e^{3x}$

   (c) $y'' - y = 2x \sin x$

2. Find the general solutions to the following non-homogeneous linear differential equations by a variation of parameters.

   (a) $y'' + 2y' + y = e^{-x} \ln x$

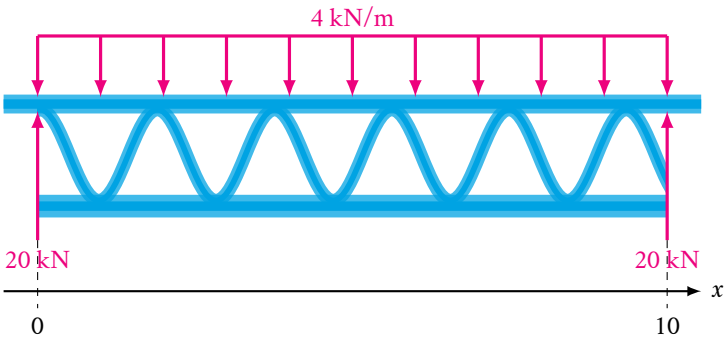
   (b) $y'' - 3y' + 2y = \sin e^{-x}$

   (c) $y'' + y = \sec x$

3. The amount of deflection (or bending) that a beam experiences can be related to the load acting on the beam through the *moment equation*

$$EIv''(x) = M(x),$$

   where $v(x)$ is the deflection and $M(x)$ is a (known) moment function. Here, E and I are constants, and the product EI represents the beam's resistance to bending. A simply supported beam of length 10 m carries a uniform load of 4 kN/m. In this case, the moment function is known to be $M(x) = 20x - 2x^2$.



   Find the function $v(x)$ for the given beam set-up. Hence, or otherwise, determine the maximum deflection that the beam experiences. At which point does this deflection occur?

Main Course

1. Consider the non-homogeneous differential equation

$$y'' + 4y = \sin^2 x.$$

   (a) Find a general solution by the method of undetermined coefficients.

   (b) Find a general solution by a variation of parameters.

   (c) Verify that the two solutions you have obtained are equivalent.

2. A ball of mass 3 kilograms extends a spring attached to the ceiling by 20 centimeters. Take the acceleration due to gravity to be $g \approx 9.8$ meters per second squared.

   (a) From rest, it is extended by an additional 30 centimeters, and then released. Find the equation of motion of the ball, and hence determine the period, frequency, and amplitude of the ball's oscillation.

   (b) A motor is attached to the spring, inducing a sinusoidal driving force $f(t) = 6\cos 7t$ Newtons on the mass-spring system. Assuming that the ball starts from rest at equilibrium, determine a function that describes the ball's motion.

3. Suppose that the motion of a damped and driven oscillator satisfies the initial value problem

$$\ddot{x} + 2\dot{x} + 2x = 4\cos t + 2\sin t, \quad x(0) = 0, \quad \dot{x}(0) = v_0.$$

   (a) Determine whether the oscillator is underdamped, critically damped, or overdamped.

   (b) Find the particular solution $x(t)$ that describes the motion of the oscillator.

   (c) Find a function that describes the oscillator's motion as t grows large: that is, find the *steady-state solution* $x_\infty(t)$.

4. Consider the differential equation

$$m\ddot{x} + kx = F_0 \cos \omega t,$$

   which models the motion of an undamped mass-spring system, externally driven by a sinusoidal force at resonant frequency.

   (a) State the value of ω. Hence, or otherwise, show that the solution to this differential equation can be written in the form

$$x(t) = R\cos(\omega t - \delta) + Ct\sin(\omega t)$$

   for some arbitrary constants R and δ. What is the value of C?

   (b) Suppose that the oscillator starts at rest at equilibrium:

$$x(0) = 0, \quad \dot{x}(0) = 0.$$

   Show that under these initial conditions, $R = 0$.

(c) A mass of 2 kg hangs from a spring with spring constant $k = 50$ N m^{-1}. It is driven from rest at equilibrium by a sinusoidal force with amplitude $F_0 = 10$ N at resonant frequency.

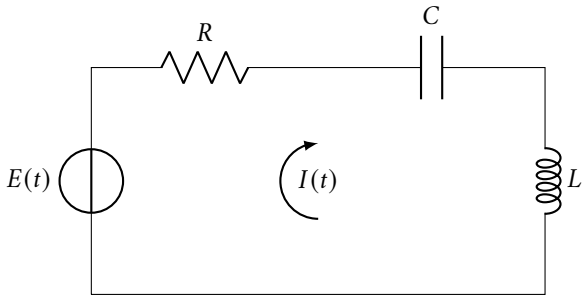
i. Determine the function $x(t)$ that describes the motion of the mass.

ii. The spring will break if the displacement of the mass ever reaches or exceeds 1 m. Estimate the time at which the spring breaks.

iii. Suppose we add a damper to the system. Find the smallest possible value of the damping constant c for which the system's amplitude at the steady-state does not exceed 1 m.

Room for Dessert

1. Consider a simple series circuit, as shown in the diagram below.



We assume that the resistance R (in ohms Ω), capacitance C (in farads F), and inductance L (in henrys H) are all positive constants, and that the impressed voltage E (in volts V) is a given function of time. By Kirchhoff's Second Law, the current I (measured in amperes A) is a function of time t that satisfies the differential equation

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = E(t), \quad \text{where} \quad I = \frac{dQ}{dt}.$$

Here, $Q(t)$ is the total charge (in coulombs C) on the capacitor at time t. If $E(t) = 0$, we say that the electrical vibrations of the circuit are free.

(a) i. Show that the above equation can be written as a second-order linear differential equation in Q.

ii. Show that the above equation can be written as a second-order linear differential equation in I.

(b) Consider a circuit with free electrical vibrations, and suppose that $C = 10^{-5}$ F, $R = 3 \times 10^2$ Ω, and $L = 0.2$ H. If the initial charge on the capacitor is 10^{-6} C and there is no initial current, find the charge Q on the capacitor at any time t.

(c) Suppose that $C = 0.8 \times 10^{-6}$ F and $L = 0.2$ H. Find the resistance R so that the circuit is critically damped.

(d) Suppose that $C = 0.25$ F, $R = 2$ Ω, $L = 1$ H, and $E(t) = 50 \cos t$ V.

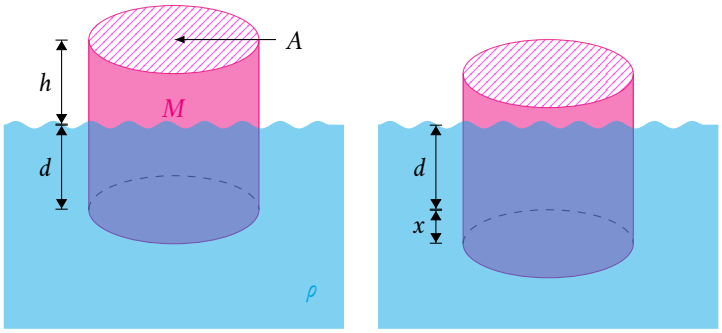
i. Determine the *steady-state* charge.

ii. Determine the *steady-state* current.

2. A fully loaded ship can be modeled as a solid object with perfectly vertical sides and a perfectly horizontal bottom, so all horizontal cross-sections have the same area, A. According to Archimedes' Principle,

   > *the upward buoyant force exerted on a body immersed in a fluid is equal to the weight of the fluid that the body displaces.*

   Let ρ be the mass density of seawater, and let M be the mass of the ship, so that its weight is Mg, where g is the acceleration due to gravity.



   (a) Find the distance d from sea level to the bottom of the ship when the ship is at rest. This is called the *draught* of the ship.

   (b) Suppose, now, that the ship is not at rest; instead, it is moving in the vertical direction. Let $d + x\,(t)$ be the distance from sea level to the bottom of the ship, where d is the draught as above. Show that if gravity and buoyancy are the only forces acting on the ship, it will bob up and down with an angular frequency given by $\omega = \sqrt{\rho A g / M}$.

   (c) Suppose, now, that waves from a storm strike the ship initially at rest. The waves exert a vertical force $F_0 \cos(\omega t)$ on the ship.

      i. What is the equation of motion of the ship?

      ii. Let h be the height of the deck of the ship above sea level when the ship is at rest. Assume that the ship is heavily loaded, so that $h \ll d$. Write down an expression that allows you to determine when the ship sinks. *There is no need to compute when the ship actually sinks.*

The Laplace Transform

Appetizers

1. Use the definition of the Laplace transform to compute $\mathcal{L}[f(t)]$ for $f(t) = e^{at}$ for some constant a. Hence, use Euler's formula to determine the Laplace transforms of $f(t) = \cos \omega t$ and $f(t) = \sin \omega t$.

2. Use the various properties of the Laplace transform to compute $\mathcal{L}[f(t)]$ for the following functions $f(t)$.

   (a) $f(t) = t^2 e^{-3t}$

   (b) $f(t) = \sin^2 t$

   (c) $f(t) = t \cos 2t$

3. For each of the following functions $f(t)$, (i) sketch the graph of $f(t)$; (ii) express $f(t)$ in terms of the unit step function; and (iii) find the Laplace transform of $f(t)$.

   (a) $f(t) = \begin{cases} 1, & 0 \le t < 5, \\ 0, & t \ge 5. \end{cases}$

   (b) $f(t) = \begin{cases} \sin t, & 0 \le t < 2\pi, \\ t - 2\pi, & t \ge 2\pi. \end{cases}$

   (c) $f(t) = \begin{cases} t, & 0 \le t < 1, \\ t^2, & 1 \le t < 2, \\ 4, & t \ge 2. \end{cases}$

4. Find the inverse Laplace transform of the following functions $F(s)$.

   (a) $F(s) = \dfrac{4}{s^2 - 1}$

   (b) $F(s) = \dfrac{s}{s^2 + 10s + 26}$

   (c) $F(s) = \dfrac{2s^3 + 4}{s^4 + 2s^3}$

5. Use the Laplace transform to solve the following initial value problem:

   $$y'' - 2y' = 4, \quad y(0) = 1, \quad y'(0) = 0.$$

   Verify that you obtain the same solution when you solve the differential equation by the method of undetermined coefficients.

Main Course

1. Let
$$f(t) = te^{at}, \quad g(t) = t\cos\omega t, \quad h(t) = t\sin\omega t.$$
   (a) Compute the Laplace transform of $f(t)$ by using the Differentiation Property.
   (b) Use Euler's formula on your answer from part (a) to determine the Laplace transforms of $g(t)$ and $h(t)$. Verify your answers by directly computing the Laplace transforms of $g(t)$ and $h(t)$ using the Differentiation Property.
   (c) Hence, or otherwise, compute the inverse Laplace transform of
$$F(s) = \frac{1}{(s^2+1)^2}.$$

2. Consider the following initial value problem.
$$y' = t \cdot u(t-2), \quad y(0) = 4.$$
   (a) Sketch the graph of the function $f(t) = t \cdot u(t-2)$.
   (b) Use the Laplace transform to solve the initial value problem. Sketch the graph of the solution $y(t)$ to the initial value problem.

3. A solid object with mass m is immersed in water with linear drag coefficient c. A pulse horizontal force of magnitude p is applied at $t = 0$: by Newton's Second Law, the equation of motion is thus given by
$$m\ddot{x} = -c\dot{x} + p \cdot \delta(t), \quad x(0) = \dot{x}(0) = 0.$$
   Solve the above initial value problem to determine the displacement $x(t)$ of the object at time t. Express your answer in terms of m, c, and p.

4. A damped oscillator at rest is excited by a unit impulse at $t = 1$ seconds. Suppose that the equation of motion is given by the second-order differential equation
$$2\ddot{x} + 2\dot{x} + 5x = \delta(t-1), \quad x(0) = \dot{x}(0) = 0.$$
   (a) Solve the initial value problem to determine the motion $x(t)$ of the oscillator. Hence, determine the time t_0 it takes for the object to complete one oscillation.
   (b) You wish to stop the oscillator at time $t = t_0$, such that it comes to rest immediately after completing one oscillation. To do so, you apply an impulse of magnitude p to the system at time $t = t_0$:
$$2\ddot{x} + 2\dot{x} + 5x = \delta(t-1) + p \cdot \delta(t-t_0), \quad x(0) = \dot{x}(0) = 0.$$
      i. Solve the modified initial value problem, expressing $x(t)$ in terms of p and t_0.
      ii. Determine the magnitude p of the impulse that should be applied to the oscillator in order to make it stop immediately.

Room for Dessert

1. Consider the function

$$f(t) = 1 + \sum_{k=1}^{\infty} (-1)^k u(t-k).$$

(a) Sketch the graph of $f(t)$.

(b) Compute the Laplace transform of $f(t)$.
Hint: Use the geometric series $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$.

(c) Solve the initial value problem

$$y' + y = f(t), \quad y(0) = 0.$$

2. Consider an undamped oscillator to which a sequence of five impulses is applied at π-second intervals:

$$\ddot{x} + x = \sum_{k=1}^{5} (-1)^{k+1} \delta(t - k\pi), \quad x(0) = \dot{x}(0) = 0.$$

(a) Without solving the differential equation, predict what the solution $x(t)$ will look like.

(b) Use the Laplace transform to solve the initial value problem. What happens to the oscillator after all five impulses have been applied?

An Introduction to Partial Differential Equations

Appetizers

1. Show that each of the following functions $u(x, y)$ satisfies the corresponding partial differential equation.

   (a) $au_x + bu_y = 0, u(x, y) = bx - ay$

   (b) $u_x + yu_y = 0, u(x, y) = e^{-x}y$

   (c) $u_x + 2xy^2u_y = 0, u(x, y) = x^2 + 1/y$

2. (a) Find a general solution $v(x)$ to the ordinary differential equation

   $$v' + 3v = 0.$$

   (b) Hence, or otherwise, find a general solution $u(x, y)$ to the partial differential equation

   $$3u_y + u_{xy} = 0.$$

   Verify that the solution you obtained indeed satisfies the above partial differential equation. *Note that your solution may involve arbitrary functions in x and y.*

   (c) What happens when you try to solve the partial differential equation using the method of separation of variables?

3. Solve the following partial differential equations by performing a separation of variables.

   (a) $yu_x - xu_y = 0$

   (b) $u_{xy} = u$

   (c) $xu_{xy} + 2yu = 0$

4. The conduction of heat on a certain metal rod is governed by the following partial differential equation:

   $$u_t = 100u_{xx}, \quad 0 < x < 1, \quad t > 0.$$

   Use the solution of the heat equation to solve the above equation under each of the following boundary and initial conditions.

   (a) $u(0, t) = u(1, t) = 0, \quad u(x, 0) = \sin 2\pi x.$

   (b) $u(0, t) = u(1, t) = 0, \quad u(x, 0) = -2 \sin 5\pi x.$

   (c) $u(0, t) = u(1, t) = 0, \quad u(x, 0) = \sin 2\pi x - 2 \sin 5\pi x.$

Main Course

1. The transport of a liquid dye dumped into a flowing river can be modeled by a partial differential equation known as the *advection equation*:

$$u_t = -\nu u_x.$$

The solution $u(x,t)$ describes the particle density of the dye at a given position x and time t, and the (non-zero) constant ν is the speed at which the river flows, carrying the dye along with it.

(a) Show that $u(x,t) = x - \nu t$ is a solution to the advection equation.

(b) Show that $u(x,t) = e^{-(x-\nu t)^2}$ is a solution to the advection equation.

(c) More generally, show that any function of the form

$$u(x,t) = f(x - \nu t)$$

is a solution to the advection equation.

2. Find a particular solution $u(x,y)$ to the partial differential equation

$$u_{xx} - 2u_x - 3u = 0,$$

given the initial conditions $u(0,y) = 2y$ and $u_x(0,y) = 2y - 4\sin y$.

3. Find a particular solution to each of the following partial differential equations by employing a separation of variables.

(a) $xu_x = u_y, u(1,0) = 9, u(2,0) = 36$

(b) $u_{xy} = u \sin y \cos x, u(\pi/2, \pi/2) = e, u(-\pi/2, -\pi/2) = e^3$

(c) $u_x + u_y = (x-y)u, u(0,0) = u(0,2) = 1$

4. Consider the following partial differential equation:

$$u_t = 2u_{xx}, \quad 0 < x < 3, \quad t > 0.$$

Solve the above equation with the given boundary and initial conditions

$$u(0,t) = u(3,t) = 0, \quad u(x,0) = \frac{5}{8}\sin \pi x - \frac{5}{16}\sin 3\pi x + \frac{1}{16}\sin 5\pi x.$$

Room for Dessert

1. The conduction of heat on a two-dimensional steel plate whose boundaries are held at thermal equilibrium is governed by the partial differential equation

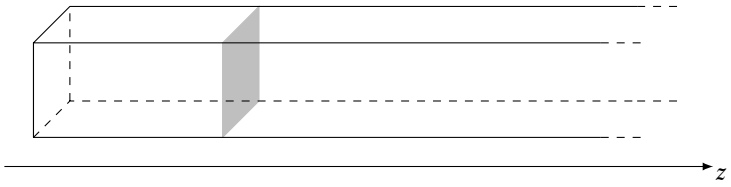
$$u_t = c^2\left(u_{xx} + u_{yy}\right).$$

Assuming that $u(x,y,t) = X(x)\,Y(y)\,T(t)$, find a set of ordinary differential equations that determine the functions $X(x)$, $Y(y)$, and $T(t)$. *Do not solve these ordinary differential equations.*

2. The heat equation was first investigated by the French mathematician and physicist Jean-Baptiste Joseph Fourier, in his groundbreaking treatise *Théorie analytique de la chaleur*, or *The Analytical Theory of Heat*, published in 1822. Fourier's most general formulation of the heat equation described the movement of heat in objects with three spatial dimensions:

$$\frac{\partial v}{\partial t} = \frac{K}{CD}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right).$$

Here, $v(x, y, z, t)$ denotes the object's temperature at position (x, y, z) and time t, while K, C, and D are some material constants.



In Chapter III, Fourier explores the dispersion of heat in an infinite rectangular solid extending in the z-direction, as illustrated above. For simplicity, Fourier assumes that the solid is uniform in the z-direction: that is, the temperature does not vary along the length of the solid.

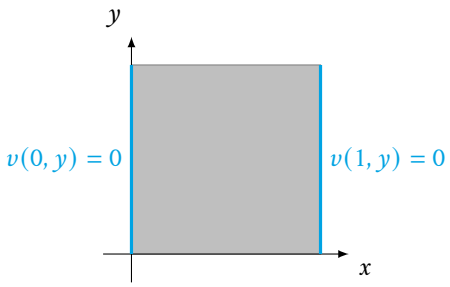
(a) After a long time, the temperature distribution of the solid approaches a *steady-state*—an equilibrium solution, independent of the variable t. With reference to Fourier's general heat equation, explain why the steady-state solution can be obtained by solving

$$v_{xx} + v_{yy} = 0.$$

This is more famously known as *Laplace's equation*.

(b) Let $v(x, y) = X(x)\,Y(y)$. Perform a separation of variables on Laplace's equation to obtain two ordinary differential equations that $X(x)$ and $Y(y)$ must satisfy.

(c) Suppose that each cross-section of the solid is a unit square, and that parallel sides of the solid are bounded by infinite blocks of ice.



This allows Fourier to impose two boundary conditions:

$$v(0, y) = 0, \quad v(1, y) = 0.$$

Determine a solution to Laplace's equation by solving the ordinary differential equations you have obtained, accounting for the given boundary conditions.

Note that only two boundary conditions have been imposed, so your final answer should still involve two arbitrary constants.

Hints and Answers to Selected Problems

Before We Begin

Calculus of a Single Variable

1. (a) $1/\sqrt{3}$. (b) 1. (c) $-\sqrt{3}/5$.

2. (a) Let $u = \sqrt{x}$, then integrate by parts.

   (b) Let $u = \ln x, du = dx/x$.

   (c) Write $\dfrac{\sqrt{1-x}}{\sqrt{1+x}} \cdot \dfrac{\sqrt{1-x}}{\sqrt{1-x}} = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{x}{\sqrt{1-x^2}}$.

   (d) Write $t \sin t \cos t = \frac{1}{2} t \sin 2t$. Let $u = 2t$, then integrate by parts.

   (e) Integrate by parts: let $u = \ln r$, $du = r^{-1}\, dr$, and $dv = r^4\, dr, v = r^5/5$.

   (f) Decompose the integrand into partial fractions.

   (g) Complete the square; then, let $u = x - 2$ and $du = dx$.

   (h) Write $\sin^5 t \cos^4 t = \left(1 - \cos^2 t\right)^2 \cdot \cos^4 t \cdot \sin t$; then, let $u = \cos t$.

   (i) Let $u = \sqrt{t}$ and $dt = 2u\, du$.

   (j) Write $\dfrac{6z+5}{2z+1} = \dfrac{3\,(2z+1)+2}{2z+1} = 3 + \dfrac{2}{2z+1}$.

The Complex Numbers

1. (a) $8 + i$. (b) $\frac{1}{13}\,(-4+7i)$. (c) $\frac{1}{5}\,(-4-7i)$. (d) $-18i$.

2. (a) $z = \sqrt{2}e^{\pi i/4}$. (b) $w = \frac{1}{2}\left(\sqrt{3} - i\right)$.

4. $z = e^{\pi/2}$.

An Introduction to Ordinary Differential Equations

Appetizers

1. To find the equilibrium solutions, find the values of y at which $y' = 0$. To determine the long-term behavior of $y(x)$, check whether y' is positive or negative for various values of $y(0)$.

2. (a) $y = \ln|x| - \ln|x+1| + c$

   (b) $y = \dfrac{\sin 6x}{12} + \dfrac{\sin 4x}{8} + c$

   (c) $-1/y + \ln|y| = x^2 - x + c$

3. $Ae^{2x} = \dfrac{x-y+2}{x-y}$; $y = x$.

4. $-\dfrac{1}{4}\ln\left|\dfrac{y}{x} - 2\right| - \dfrac{3}{4}\ln\left|\dfrac{y}{x} + 2\right| + c = \ln|x|$; $y = \pm 2x$.

5. Approximately 40% of C_i.

6. About 14641; about 18.18 hours.

Main Course

1. (a) Let $u = x + y$; $x + y + \ln\left|(x+y)^2 + 4x + 4y + 5\right| = 2x + \ln 5$.

   (b) Let $u = y/x$; $\ln x = -e^{-y/x} + 1$.

   (c) Let $u = y'$; $y = \dfrac{4}{3}(x+1)^{3/2} - \dfrac{1}{3}$.

2. (b) Let $u = dy/dx$; $y(x) = \dfrac{T}{\mu}\left(\cosh\left(\dfrac{\mu}{T}x\right) - 1\right)$.

3. (a) $y'(t) = 0.75 - 0.03y$. (b) $y(t) = 25$ is a stable equilibrium.

4. (a) About 140 bugs per day. (b) About 311 bugs.

Linear Differential Equations

Appetizers

1. (a) Integrating factor xe^x; $xe^x y = x + c$.

   (b) Integrating factor $e^{\sin x}$; $e^{\sin x} y = x + c$.

   (c) Integrating factor $x^{-3}e^{-x}$; $x^{-3}e^{-x}y + c = -e^{-x}x^{-2}$.

2. (a) $y = 3 - x$.

   (b) $y = e^{-3x}\left(1 + 2x\right)$.

   (c) $y = e^{x}\left(-2\cos 2\pi x + 3\sin 2\pi x\right)$.

3. $\theta\left(t\right) = \left(1/\omega\right)\sin \omega t$; approximately 0.19 radians.

Main Course

1. (a) Let $v = y^2$; $\dfrac{e^x}{x}y^2 = \dfrac{1}{2}e^{2x} + c$.

   (b) Let $v = y^2$; $e^{2x}y^2 = -xe^{2x} + \dfrac{1}{2}e^{2x} + c$.

   (c) Let $v = \sin y$; $e^{-x}\sin y = -x^2e^{-x} - 2xe^{-x} - 2e^{-x} + c$.

2. (a) $T'\left(t\right) = -k\left(T\left(t\right) - T_{\text{env}}\right)$, where $k > 0$.

   (b) Integrating factor e^{kt}; $T\left(t\right) = T_{\text{env}} + ce^{-kt}$.

   (c) (i) About 94°C. (ii) About 3 hours.

3. (a) $y = c_1e^{-x} + c_2e^{x} + c_3xe^{x}$.

   (b) $y = c_1e^{-3x} + c_2e^{-2x} + c_3e^{2x} + c_4e^{3x}$.

   (c) $y = c_1e^{\sqrt{2}x} + c_2xe^{\sqrt{2}x} + c_3e^{-\sqrt{2}x} + c_4xe^{-\sqrt{2}x}$
   $+c_5\cos\sqrt{2}x + c_6\sin\sqrt{2}x + c_7x\cos\sqrt{2}x + c_8x\sin\sqrt{2}x$.

4. (b) i. $m\ddot{y} = -ky$. ii. $\omega = \sqrt{k/m}$. iii. $0 < c < 2\sqrt{mk}$.

The Harmonic Oscillator

Appetizers

1. (a) $y = e^{-x}\left(c_1 \cos 3x + c_2 \sin 3x\right) + \frac{5}{2}x^2 - x$.

   (b) $y = c_1 e^{4x} + c_2 e^{2x} - \left(x^2 + 2\right) e^{3x}$.

   (c) $y = c_1 e^{-x} + c_2 e^{x} - \cos x - x \sin x$.

2. (a) $y = c_1 e^{-x} + c_2 x e^{-x} + \left(\frac{1}{2}\ln x - \frac{3}{4}\right) x^2 e^{-x}$.

   (b) $y = c_1 e^{x} + c_2 e^{2x} - e^{2x} \sin e^{-x}$.

   (c) $y = c_1 \cos x + c_2 \sin x + \ln|\cos x| \cos x + x \sin x$.

3. $v(x) = \frac{1}{EI}\left(\frac{10}{3}x^3 - \frac{1}{6}x^4 - \frac{500}{3}x\right)$.

   The maximum deflection occurs at $x = 5$: $v(5) = -3125/6EI$.

Main Course

1. $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{8} - \frac{1}{8}x \sin 2x$.

2. (a) $y(t) = 0.3 \cos 7t$; $\tau = 2\pi/7$; $f = 7/2\pi$; amplitude is 0.3 m.

   (b) $y(t) = (1/7)\, t \sin 7t$.

3. (a) The system is underdamped.

   (b) $x(t) = (v_0 - 2)\, e^{-t} \sin t + 2 \sin t$.

   (c) $x_\infty(t) = 2 \sin t$.

4. (a) $\omega = \sqrt{k/m}$; $C = F_0/2m\omega$.

   (c) (i) $x(t) = 0.5t \sin 5t$. (ii) $t \approx 2.13$ s. (iii) $c = 2$ N s m^{-1}.

The Laplace Transform

Appetizers

1. Let $a = i\omega$; separate real and imaginary parts.

2. (a) Use the First Shifting Theorem; $\mathcal{L}[f(t)] = \dfrac{2}{(s+3)^3}$.

   (b) Rewrite $\sin^2 t$, or use the transform of derivatives;
   $\mathcal{L}[f(t)] = \dfrac{2}{s(s^2+4)}$.

   (c) Use the Differentiation Property; $\mathcal{L}[f(t)] = \dfrac{s^2-4}{(s^2+4)^2}$.

3. (a) $f(t) = 1 - u(t-5)$; $\mathcal{L}[f(t)] = \dfrac{1}{s} - \dfrac{e^{-5s}}{s}$.

   (b) $f(t) = \sin t + (t - 2\pi - \sin t) \cdot u(t-2\pi)$;
   $\mathcal{L}[f(t)] = \dfrac{1}{s^2+1} + e^{-2\pi s}\left(\dfrac{1}{s^2} - \dfrac{1}{s^2+1}\right)$.

   (c) $f(t) = t + t(t-1) \cdot u(t-1) + \left(4 - t^2\right) u(t-2)$;
   $\mathcal{L}[f(t)] = \dfrac{1}{s^2} + \dfrac{(s+2)e^{-s}}{s^3} - e^{-2s}\left(\dfrac{2}{s^3} + \dfrac{4}{s^2}\right)$.

4. (a) Decompose into partial fractions; $\mathcal{L}^{-1}[F(s)] = 2e^t - 2e^{-t}$.

   (b) Complete the square; $\mathcal{L}^{-1}[F(s)] = e^{-5t}(\cos t - 5\sin t)$.

   (c) Decompose into partial fractions; $\mathcal{L}^{-1}[F(s)] = \dfrac{1}{2} - t + t^2 + \dfrac{3}{2}e^{-2t}$.

5. $y = -2t + e^{2t}$.

Main Course

1. (a) $\mathcal{L}[f(t)] = \dfrac{1}{(s-a)^2}$.

   (b) $\mathcal{L}[g(t)] = \dfrac{s^2-\omega^2}{(s^2+\omega^2)^2}$, $\mathcal{L}[h(t)] = \dfrac{2s\omega}{(s^2+\omega^2)^2}$.

   (c) $\mathcal{L}^{-1}[F(s)] = \dfrac{1}{2}(\sin t - t\cos t)$.

2. (b) $y(t) = \left(\dfrac{t^2}{2} - 2\right) \cdot u(t-2) + 4$.

3. $x(t) = \dfrac{p}{c}\left(1 - e^{-(c/m)t}\right)$.

4. (a) $x(t) = \dfrac{1}{3}\left[e^{-(t-1)/2}\sin\left(\dfrac{3}{2}(t-1)\right)\right] \cdot u(t-1)$; $t_0 \approx 5.19$ seconds.

   (b) i. $x(t) = \dfrac{1}{3}\left[e^{-(t-1)/2}\sin\left(\dfrac{3}{2}(t-1)\right)\right] \cdot u(t-1)$
   $+\dfrac{p}{3}\left[e^{-(t-t_0)/2}\sin\left(\dfrac{3}{2}(t-t_0)\right)\right] \cdot u(t-t_0)$.

   ii. Find the value of p such that the amplitudes of the oscillations in $x(t)$ cancel each other out at $t = t_0$; $p \approx -0.123$.

An Introduction to Partial Differential Equations

Appetizers

1. Compute the partial derivatives for each function $u(x, y)$, and plug them into the corresponding partial differential equation.

2. (a) $v(x) = ce^{-3x}$.
   (b) $u(x, y) = e^{-3x} f(y) + g(x)$.

3. (a) $u(x, y) = Ce^{c\left(x^2+y^2\right)}$.
   (b) $u(x, y) = Ce^{kx+y/k}$.
   (c) $u(x, y) = Cx^k e^{-y^2/k}$.

4. (a) $u(x, t) = e^{-400\pi^2 t} \sin(2\pi x)$.
   (b) $u(x, t) = -2e^{-2500\pi^2 t} \sin(5\pi x)$.
   (c) $u(x, t) = e^{-400\pi^2 t} \sin(2\pi x) - 2e^{-2500\pi^2 t} \sin(5\pi x)$.

Main Course

1. Compute the partial derivatives for each function $u(x, t)$, and plug them into the advection equation.

2. Solve an ordinary differential equation in x, and let the constants be arbitrary functions of y; $u(x, y) = (y + \sin y) e^{-x} + (y - \sin y) e^{3x}$.

3. (a) $u(x, y) = 9x^2 e^{2y}$.
   (b) $u(x, y) = e^{2-\sin x+\cos y}$.
   (c) $u(x, y) = e^{\frac{1}{2}\left(x^2-y^2\right)-x+y}$.

4. This is the heat equation with $c^2 = 2$ and $\ell = 3$;

$$u(x, t) = \frac{5}{8} e^{-2\pi^2 t} \sin(\pi x) - \frac{5}{16} e^{-18\pi^2 t} \sin(3\pi x) + \frac{1}{16} e^{-50\pi^2 t} \sin(5\pi x).$$

You've made it to the end!



Illustration of a vintage nobleman, courtesy of Wikimedia Commons.